Sensors For
DEEP-SEA APPLICATIONS

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This article covers some important sensors used for different deep-sea applications, including civil, military, research, communication, seismic and energy

Mysteries of the deep sea, vastness of the blue waters stretching out to the horizon, and the tremendous potential of the oceans for military and civil applications have let humans explore the world of the mighty waters—an effort that is still ongoing.

Deep-sea exploration gives mankind a sense of human progress, because it involves great manpower and technology. It also leads to academic and industry researchers show great interest in deep-sea sensors. There is good progress being made in modern deep-sea research and development with recent advancements in sensors, microelectronics and computers.

Modern electronic instruments in oceanography have helped humans explore further deep down the ocean and observe the greatest depths. Electronic systems with low power, accurate and delicate sensors collect vital information, which has helped researchers measure and record the deep-sea environment and tidal data.

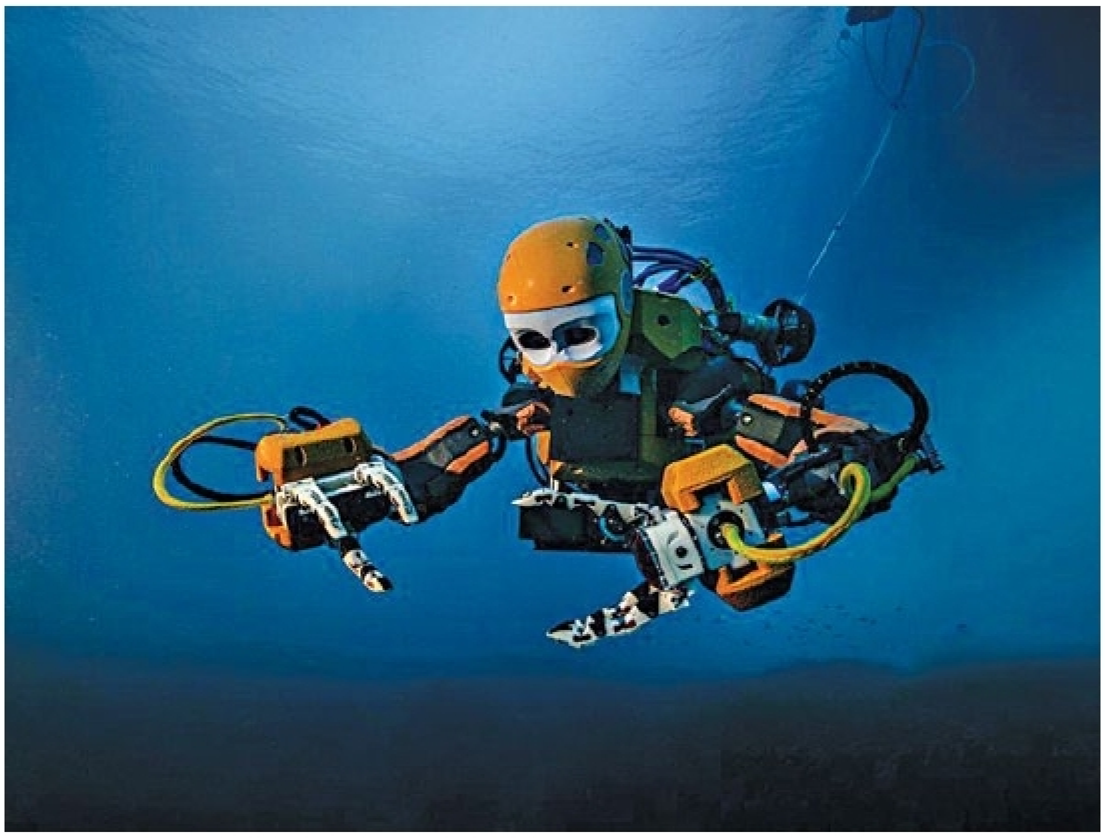

Fig. 1: Robotic mermaid with touch sensors (Credit: www.livescience.com)

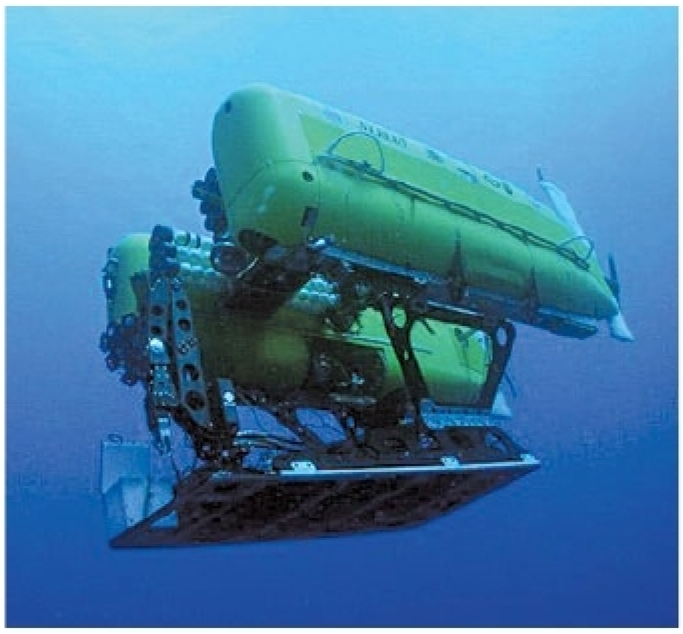

Fig. 2: Nereus, the world's deepest diving underwater vehicle (Credit: www.bbc.com)

Underwater drones and portable submersible robots with intelligent sensors for monitoring water temperature, auto compass heading and depth, turns, pitch and roll have been developed. Due to technological advancements, underwater drones can now move around freely, faster and more easily.

Robots and artificial intelligence (AI) have been developed to explore the oceans where humans cannot reach. Stanford University has developed a robotic mermaid named OceanOne to dive deeper than humans can. This robot's human-like body with touch sensors can carry out a variety of tasks in the deep sea. Its wrists are fitted with force sensors that give haptic feedback to the operator's control. These feedback signals can identify actions such as the robot grasping at something firm and heavy, or light and delicate. The brain of the robot reads data and makes sure that its hands keep a firm grip on objects.

Nereus is the world's deepest-diving underwater vehicle. It can be configured to operate as a remotely-operated vehicle (ROV), or operate independently of human control as an autonomous underwater vehicle (AUV). Nereus is capable of explor-